

Amandeep Srivastava

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SUMMARY

Infrastructure and distributed-systems engineer with **7+ years** of experience building control planes and developer platforms. Architected the greenfield **Infrastructure-as-Code authoring platform** for a control plane used by **10,000+ engineers** to manage **~8M resources** across 10+ platforms, and drove zero-outage migrations across **90+ teams**.

TECHNICAL SKILLS

Languages: Java, Go, Python, C++

Distributed Systems & Infrastructure: Kubernetes, Crossplane, Helm, gRPC

Data & Streaming: Kafka, Samza, Cassandra, Vespa, Elasticsearch, MySQL

Graph & Backend: Gremlin, SPARQL, JanusGraph, AWS Neptune, Spring Boot, Vert.x

AI Systems: Agentic platform tooling and autonomous workflows (Claude SDK, MCP, GitHub Copilot SDK)

EXPERIENCE

LinkedIn | Senior Software Engineer

Sep 2024 – Present

Core Infrastructure, Control Plane • Tech lead: authoring, CI, and AI pillars of the IaC initiative

- Architected LinkedIn's **Infrastructure-as-Code authoring platform** and unified CLI, now the standard authoring path for **5,000+ services**; shift-left validation moves misconfiguration detection from deploy time to PR time, while multithreading and rule-engine tuning cut validation runtime **~90%**, together yielding **~\$5M/year** in projected savings.
- Led a team of five engineers to design and deliver a fully backward-compatible, multi-file manifest model that lets one manifest span repositories, enabling file-level resource access control and bounded automation context; drove alignment across partner teams and landed it on **154 services across 90+ teams with zero outages**.
- Engineered **Long-Running Operations** for the **Kubernetes/Crossplane** provisioning engine, modeling multi-step operations as resumable Kubernetes-object state machines and cutting provisioning time **~73%**; the model is now being extended to stateful cluster management.
- Built **LinkedIn's first infrastructure-management agent**, converting natural-language requests into validated infrastructure from the IDE and CLI through a pluggable rule model; a separate autonomous migration agent onboarded **250+ pre-IaC services** with zero outages, cutting per-service effort from **~2 hours** to **~20 minutes**.
- Productionized an **agentic PR-review service** for IaC changes, now reviewing **100% of migration PRs**, and a **developer on-call agent** that triages across design docs, wikis, Slack, tickets, and code.
- Set technical direction across the authoring, CI, and AI pillars: authored the design documents partner teams build against, defined the authoring and validation standards platforms implement, owned design and code review approval, mentored **8+ engineers**, and conducted **100+ technical interviews**.

LinkedIn | Software Engineer

May 2022 – Sep 2024

Systems Infrastructure

- Developed the resource-manager core of a new, centrally managed **infrastructure control plane** and onboarded **MySQL** as its first tenant as the sole engineer, delivering self-serve provisioning with 70% fewer errors; **the pattern established there is now reused across 30+ platforms**, including Couchbase, Temporal, and TiDB.
- Designed and shipped, solo and end-to-end, a high-throughput (**~30M QPS**) **resource-insights service** accounting for **~17%** of platform API traffic; it became a core dependency of database provisioning, quota changes, deletion safety checks, and infrastructure-cost reporting.

Goldman Sachs | Software Engineer → Senior Software Engineer

May 2019 – May 2022

Knowledge Management & Research team

- Built a **database-agnostic graph platform** (AWS Neptune, JanusGraph backed by a **Cassandra** store, Gremlin) for fraud detection across firm transaction records, with an **Elasticsearch**-backed search service for autocomplete, fuzzy, and weighted entity lookups over the graph.
- Replaced a legacy **Attivio** (Lucene-based) search stack with a **Vespa** document search store, cutting response times by **~35%**.

EDUCATION

IIT (ISM) Dhanbad

B.Tech (Hons.), Computer Science and Engineering

2015 – 2019

CGPA: 9.24 / 10

RECOGNITION

Global winner - Core Infrastructure AI Hackathon @ LinkedIn; Google Summer of Code Mentor, DBpedia (2019 & 2020); ranked **1 of 950+** in first-semester examinations at IIT (ISM) Dhanbad.